



CONTACT & MEDIA

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in/dtanajewski

KEY SKILLS & EXPERTISE AREAS

Earth Observation / EO

Machine learning & AI

Geophysics / GPR

GNSS & navigation

R&D project management

LANGUAGES

Polish, Native

English, B2

SOFTWARE

Python (rasterio, GeoPandas, sklearn)

QGIS · ArcGIS · SNAP

SentinelHub · Git

Dariusz Tanajewski, PhD

HEAD OF SCIENCE & INNOVATION

SUMMARY

Geospatial data science leader with 15+ years of experience, including 5+ years focused on Earth Observation, bridging EO science and industrial deployment. As a Geospatial Data Scientist at TerraEye, I lead the development of geospatial intelligence solutions that support the mining industry across the full project lifecycle—from exploration to reclamation. With a PhD in technical sciences, an academic background in computer science, and postgraduate studies in Data Science, I combine theoretical rigour with practical application of satellite data analysis and artificial intelligence. My work focuses on integrating multisensor data and applying advanced geospatial analytics and machine learning to mineral exploration and geological interpretation.

EXPERIENCE

- 05.2026 – now

Head of Science and Innovation

TerraEye, Wrocław

- Lead the Science team and development of geospatial intelligence solutions for the mining industry

- 05.2024 – 04.2026

Geospatial Data Scientist

TerraEye, Wrocław

- Develop and validate remote-sensing methodologies (Sentinel-1/2, Landsat, EMIT, thermal) for mineral exploration
- Integrate multisensor data with ML and knowledge-based reasoning for prospectivity mapping; oversee methodology rigour

R&D Projects:

EXPLOMAT. Sustainable Exploration Technologies for Raw Materials Discovery,

GRENMIN. Gravitational Energy Storage in the post-Mine areas,

RISPECT. Regional Innovation Scheme Project for Exploration Capabilities and Technologies.

TRAININGS

AI4EO Course; ML Challenge;

Digital Image Processing
with OpenCV in Python

Writing in the Sciences,
Stanford

Introduction to Computer
Science and Programming
Using Python

Git; ArcGIS; GPR
interpretation

CERTIFICATIONS

Driving licence, 2005

Sailing licence, 2018

Motorboat licence, 2019

OTHER

Member, Polish Space
Industry Assoc.

Member, IEEE Geoscience
and Remote Sensing Society

ORCID 0000-0002-9734-1762

- 05.2022 – 04.2024

R&D Team Leader

Remote Sensing Environmental Solutions, Wrocław

- Led an R&D team developing satellite-based AI solutions for the mining and environmental sector
- Managed or participated in NCBR-funded R&D projects (incl. HYPE4EXPLOR, ECOSAT, MINECAM) – from methodology design to delivery
- Directed development of mineral-prospecting and mine-monitoring tools using EO data and machine learning

- 08.2021 – 04.2022

Machine Learning Engineer

Four Point, Wrocław

- Developed machine learning models for satellite-image analysis in the New Space sector
- Applied computer vision and deep learning to Earth Observation data

- 10.2016 – 09.2022

Assistant Professor (also Adjunct Lecturer 2021–2022)

University of Warmia and Mazury in Olsztyn, Poland

- R&D Project
GPR-based detection and corrosion assessment of steel bars in reinforced concrete (as R&D team leader). Funded by the National Science Centre;
- Geolocation Support for Outdoor Tourism Project & Global Yacht Positioning System. Both were funded by ESA
- GPR surveys as team leader (planning, measurements, interpretation)
- Polish-Georgian Expedition, Kutaisi Archaeological Landscape, Georgia

- 11.2014 – 12.2021

Geophysicist & GIS / Data Analyst

GeoScience (owner) · Visimind Ltd · Environmental Expertise Rakus

- GPR acquisition and interpretation; geodetic and environmental GIS analysis
- Data analysis at Visimind Ltd (10.2015–02.2016)

EDUCATION

- 10.2011 – 11.2017
PhD in Technical Sciences
University of Warmia and Mazury in Olsztyn, Poland
Faculty of Geodesy, Spatial Engineering and Construction
- 10.2020 – 06.2022
MSc studies in Computer Science
University of Warmia and Mazury in Olsztyn, Poland
Systems and Computer Networks Design
- 2018 – 03.2019
Postgraduate Diploma – Data Science in Practice
University of Warmia and Mazury in Olsztyn, Poland
Faculty of Mathematics and Computer Science
- 2006 – 07.2011
MSc Eng., Geodesy and Satellite Navigation
University of Warmia and Mazury in Olsztyn, Poland
Research internship: Kauno Kolegija, Kaunas, Lithuania (2015)

PUBLICATIONS

- 2026
Uncertainty-Weighted Spectral Similarity for Hyperspectral Imaging Spectroscopy
Tanajewski, Jabłońska, Jakubowicz. IEEE Geoscience and Remote Sensing Letters.
doi: 10.1109/LGRS.2026.3695563
- 2025
Metric Learning Networks for Accurate Mineral Detection with different augmentation techniques
Jabłońska, Zięba, Tanajewski. Expert Systems with Applications, 130692
doi:10.1016/j.eswa.2025.130692
- 2024
MineCam: Combined Remote Sensing and Machine Learning for Segmentation and Change Detection of Mining Areas
Jabłońska, Maksymowicz, Tanajewski et al. Remote Sensing, 16, 955
- 2024
Optimal GPS/EGNOS Positioning Model Using Smartphone
Grunwald, Ciećko, Krasuski, Tanajewski. Applied Sciences, 14, 1840
- 2022
Conceptual framework of a Global Yacht Positioning System in Poland
Kaźmierczak, Szczepańska, Grunwald, Kowalczyk, Tanajewski. Applied Geomatics, 14, 79–91
Awards: Geospatial World 50 Rising Stars 2026; ESNC Special Award 2015 (German Aerospace Center (DLR))
- 2020
Theoretical Concept for a Mobile Underwater Radio-Navigation System Using Pseudolite Buoys
Grosch, Enneking, Greda, Tanajewski, Grunwald, Ciećko. Remote Sensing, 12(21)
- 2016
Application of GPR and GPS surveys for monitoring levees and dykes
Tanajewski, Bakuła. Acta Geophysica, 64(4), 1093–1111